

Code-Layout, Readability and Reusability

ComicCraft - AI Comic Story Creator using Gemini Models
Madras Institute of Technology | Department of Computer Technology (CT) | III Year, V Semester
Project period: 1 Oct 2026 - 2 Oct 2026

Layout

Path	Role
app/main.py	Entry point
app/routes.py	HTTP layer and orchestration
app/gemini_flash.py, gemini_pro.py	Text AI adapters
app/image_generator.py	Image AI adapter
app/layout_builder.py	Pure data merge (no I/O)
app/exporters.py	PDF output
templates/, static/	Presentation

Readability

- One responsibility per module; each file under 80 lines
- Type hints and short docstrings on public functions
- Constants (MODEL, PANELS, MODEL_ID) at the top of each module
- Secrets and model names read from environment, never hard-coded

Reusability

- create_comic() is shared by the HTML and JSON routes
- build_comic_layout is pure and unit-testable without any API
- Model names can be swapped through .env without code changes
- generate_image can be reused for the /test-image route

Error handling

Pipeline failures are caught in create_comic() and returned as HTTP 500 with a readable message, for example the Gemini quota error.